

Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel Level 3 GCE

Friday 17 May 2024

Afternoon

Paper reference **8FM0/23**

Further Mathematics
Advanced Subsidiary
Further Mathematics options
23: Further Statistics 1
(Part of options B, E, F and G)

You must have:
 Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Values from statistical tables should be quoted in full. If a calculator is used instead of tables the value should be given to an equivalent degree of accuracy.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. Sharma believes that each computer game he sells appeals equally to all age ranges. To investigate this, he takes a random sample of 100 people who play these games and asks them which of the games A , B or C they prefer. The results are summarised in the table below.

Computer game		A	B	C
Age range	< 20	8	15	6
	$20 - 30$	21	12	9
	> 30	6	10	13

- (a) Write down hypotheses for a suitable test to assess Sharma's belief. (1)
- (b) For the test, calculate the expected frequency for
 - (i) those players aged under 20 who prefer game C
 - (ii) those players aged between 20 and 30 who prefer game A(2)
- (c) State the degrees of freedom of the test statistic for this test. (1)

Sharma correctly calculates the test statistic for this test to be 11.542 (to 3 decimal places).

- (d) Using a 5% significance level, and stating your critical value, comment on Sharma's belief. (2)



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Question 1 continued

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Question 1 continued

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Question 1 continued

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(Total for Question 1 is 6 marks)



2. A manager keeps a record of accidents in a canteen.

Accidents occur randomly with an average of 2.7 per month. The manager decides to model the number of accidents with a Poisson distribution.

- (a) Give a reason why a Poisson distribution could be a suitable model in this situation. (1)
- (b) Assuming that a Poisson model is suitable, find the probability of
- (i) at least 3 accidents in the next month, (1)
- (ii) no more than 10 accidents in a 3-month period, (2)
- (iii) at least 2 months with no accidents in an 8-month period. (4)

One day, two members of staff bump into each other in the canteen and each report the accident to the manager. The canteen manager is unsure whether to record this as one or two accidents.

Given that the manager still wants to model the number of accidents per month with a Poisson distribution,

- (c) state
- a property of the Poisson distribution that the manager should consider when deciding how to record this situation
 - whether the manager should record this as one or two accidents
- (1)

The manager introduces some new procedures to try and reduce the average number of accidents per month.

During the following 12 months the total number of accidents is 22
The manager claims that the accident rate has been reduced.

- (d) Use a 5% level of significance to carry out a suitable test to assess the manager's claim.
You should state your hypotheses clearly and the p -value used in your test. (4)



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Question 2 continued

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Question 2 continued

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Question 2 continued

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(Total for Question 2 is 13 marks)



3. The discrete random variable X has probability distribution,

x	-1	0	1	3	7
$P(X=x)$	p	r	p	0.3	r

where p and r are probabilities.

Given that $E(X) = 1.95$

find the exact value of $E(\sqrt{X+1})$ giving your answer in the form $a+b\sqrt{2}$ where a and b are rational.

(6)



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Question 3 continued

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(Total for Question 3 is 6 marks)



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Question 4 continued

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Question 4 continued

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Question 4 continued

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Question 4 continued

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(Total for Question 4 is 15 marks)

TOTAL FOR FURTHER STATISTICS 1 IS 40 MARKS

